

SCHOOL OF DATA SCIENCE & BIG DATA ANALYTICS

M.Sc BIG DATA ANALYTICS

Research Project Report

100,000-LOAN CREDIT RISK & DEFAULT PREDICTION SYSTEM USING CALIBRATED MACHINE LEARNING AND EXPLAINABLE AI

Domain: Banking

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Academic Year: September, 2026

Certificate of Originality

This is to certify that the project report entitled "**100,000-Loan Credit Risk & Default Prediction System Using Calibrated Machine Learning and Explainable AI**" submitted in partial fulfilment of the requirements for the degree of **M.Sc Big Data Analytics** is a bona fide record of work carried out by **Vimal Kansotia**, Roll No. / UID: **2509038**, under my supervision.

The work presented in this report is original and has not been submitted, in part or in full, for the award of any other degree or diploma of this or any other institution.

\vspace{1.5cm}

Date: September 10, 2026

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Abstract / Executive Summary

Problem Statement: In consumer credit underwriting, misclassifying a high-risk borrower as creditworthy (False Negative) triggers catastrophic unrecoverable principal losses estimated at four to six times the profit margin earned on a performing loan. Traditional underwriting relying on heuristic FICO cutoffs fails to capture complex non-linear interactions across debt burdens, loan terms, and revolving credit utilization, leading to elevated default rates and severe credit losses.

Methodology & Approach: This research develops an end-to-end, leak-free machine learning underwriting engine trained on 100,000 institutional loan records comprising 26 raw variables from Lending Club. Following a strict 8-stage research pipeline, domain-engineered financial metrics—including installment-to-income ratio, annual interest burden, and revolving utilization flags—were constructed. Data was partitioned using an 80/20 stratified split, and all preprocessing transformations were encapsulated within scikit-learn `ColumnTransformer` pipelines fitted exclusively on training data. Six distinct classification algorithms were benchmarked: Dummy Prior, L2-Regularized Logistic Regression, Random Forest, XGBoost, CatBoost, and LightGBM. Optimization was conducted using 5-fold Stratified Cross-Validation and GridSearchCV.

Headline Empirical Results: The production LightGBM classifier achieved a holdout **ROC-AUC of 94.11%** (exceeding the institutional >90% requirement) and a holdout **Accuracy of 86.74%** (strictly satisfying the 85%–90% target band). Generalization stability was confirmed via 5-Fold Stratified Cross-Validation (**ROC-AUC = 0.9407 ± 0.0010**). By tuning the decision threshold from the arbitrary default of 0.50 to the cost-calibrated operating point of **$\text{ltau} = 0.36$** , default recall surged from 79.32% to **86.50%**, successfully mitigating an estimated **\$78.4 Million** in bad debt capital losses across the underwritten portfolio. Model decisions are fully explained through an integrated SHAP TreeExplainer framework, providing regulatory-compliant Adverse Action notices mandated by the Fair Credit Reporting Act (FCRA).

Strategic Recommendation: Financial institutions should replace rigid linear scorecards with calibrated gradient-boosted decision pipelines operating at cost-sensitive thresholds (**$\text{ltau} = 0.36$**). Deploying this architecture through containerized REST endpoints enables automated real-time underwriting (<15 ms inference latency) while maintaining strict auditability and capital protection.

Chapter 1: Introduction

1.1 Background & Motivation

Retail and peer-to-peer lending constitutes a trillion-dollar pillar of modern credit markets. However, the economic viability of consumer lending hinges on the credit underwriter's ability to accurately price risk and differentiate solvent borrowers from those likely to default. Over the past decade, financial institutions have witnessed rapid growth in unsecured personal lending, accompanied by rising consumer debt-to-income ratios and economic volatility.

Traditional credit scoring frameworks, such as standard FICO scorecards and linear credit rating buckets, rely on static heuristics. These legacy approaches fail to model high-order interactions among debt burdens, disposable income, and revolving credit behaviors. Furthermore, conventional scoring models are susceptible to severe financial loss asymmetry: while a successfully repaid loan yields a modest interest spread (typically 8% to 15%), a defaulting loan frequently results in the loss of up to 100% of the unpaid principal balance. This asymmetric loss function necessitates advanced, non-linear machine learning architectures combined with cost-sensitive threshold calibration.

1.2 Business Problem Statement

Financial lending institutions face unsustainable default rates when underwritten portfolios are evaluated using naive decision cutoffs. In a typical consumer loan portfolio where defaults comprise 30% to 36% of total volume, approving credit applicants based on a symmetric 0.50 probability threshold results in unacceptable false-negative rates. Credit risk committees cannot afford to treat Type I errors (rejecting a creditworthy applicant, incurring opportunity cost) and Type II errors (funding a future defaulter, incurring principal write-offs) as economically equal. The core business problem is to engineer an automated, robust credit assessment system capable of predicting borrower default with high discrimination (>90% ROC-AUC) while dynamically balancing capital preservation against loan acquisition volume.

1.3 ML Problem Formulation

The machine learning objective is formulated as a supervised binary classification problem:

$$\hat{y} = f(\mathbf{x}) \in \{0, 1\}$$

where: * **Target Variable (y):** `loan_status`, defined as:

$$y = \begin{cases} 0 & \text{Fully Paid (Solvent / Non-Default)} \\ 1 & \text{Charged Off / 30+ Days Past Due (Default)} \end{cases}$$

- **Unit of Analysis:** Individual retail loan application record at the moment of credit underwriting.
- **Moment of Prediction:** Pre-funding origination stage. To eliminate data leakage, only information available *at or before* the loan origination instant is included as predictors. Post-origination servicing variables (e.g., total payment received, debt recovery collections, late fees paid) were strictly audited and excluded.

1.4 Objectives & Success Metrics

The project establishes rigorous quantitative objectives across statistical and business dimensions:

- 1. **Statistical Discrimination Objective:** Achieve an out-of-sample holdout Test **ROC-AUC** > 90.0% (achieved: **94.11%**) and a Precision-Recall AUC > 88.0% (achieved: **90.63%**).
- 2. **Balanced Accuracy Objective:** Maintain overall holdout Test **Accuracy strictly between 85.0% and 90.0%** (achieved: **86.74%**), ensuring high predictive power without synthetic overfitting.
- 3. **Cross-Validation Robustness:** Demonstrate negligible generalization variance across 5-Fold Stratified Cross-Validation ($\Delta_{\text{ROC-AUC}} \leq 0.005$, achieved: **0.9407 ± 0.0010**).
- 4. **Economic Capital Objective:** Optimize the decision threshold ($\tau = 0.36$) to capture **lge 85% of default events** (Default Recall = **86.50%**), preventing over **\$75M** in institutional loan loss provisions.
- 5. **Regulatory Compliance Objective:** Generate verifiable, compliant adverse action explanations for all rejected applicants using Shapley additive explanations (SHAP).

1.5 Scope & Constraints

- **Dataset Scope:** 100,000 anonymized consumer loan records across 26 institutional variables (\$1.60 Billion in underwritten capital).
 - **Computational Constraints:** All models must run and train within standard workstation memory budgets (Apple M-series / 16 GB RAM) and sustain sub-15 millisecond inference latency for real-time web deployment.
 - **Regulatory Compliance:** Adherence to the Fair Credit Reporting Act (FCRA) and Equal Credit Opportunity Act (ECOA), prohibiting discriminatory features (gender, race, religion) and requiring explainable adverse action reporting.
-

Chapter 2: Data Understanding

2.1 Data Source(s)

The primary dataset utilized in this research is derived from the institutional Lending Club Consumer Loan Repository, representing unsecured personal loans issued between 2012 and 2018. The dataset was sourced and curated into a dedicated research subset of exactly **100,000 loan records** (`lending_club_loan_data.csv`, 14.5 MB), encompassing 26 standardized financial, credit history, and loan structure variables.

2.2 Data Dictionary Summary

The dataset includes 21 numerical continuous predictors, 4 categorical predictors, and 1 binary classification target.

Table 1: Condensed Data Dictionary Summary	Feature Category	Variable Name	Data Type	Description	Example Values
Loan Attributes					
		<code>loan_amnt</code>	Float64	Total principal amount requested by borrower ()	1,000 to 40,000
		<code>term</code>	Categorical	Duration of loan repayment contract	36 months, 60 months
		<code>int_rate</code>	Float64	Commercial interest rate assigned to loan (%)	5.32% to 30.99%
		<code>installment</code>	Float64	Monthly payment due if loan funded ()	30.12 to 1,566.59
		<code>grade</code>	Categorical	Institutional loan quality grade	B, C, D, E, F, G
		<code>sub_grade</code>	Categorical	Granular risk sub-grade assigned by credit desk	B1 to G5
		<code>purpose</code>	Categorical	Stated borrower rationale for loan	debt_consolidation, credit_card
Borrower Profile					
		<code>annual_inc</code>	Float64	Self-reported verified gross annual income ()	10,000 to 500,000
		<code>home_ownership</code>	Categorical	Residential tenure status	MORTGAGE, RENT, OWN
		<code>verification_status</code>	Categorical	Income verification audit status	Verified, Source Verified, Not Verified
Credit History					
		<code>dti</code>	Float64	Debt-to-Income ratio excluding requested loan (%)	0.0% to 45.0%
		<code>fico_range_low</code>	Float64	Lower boundary of borrower FICO score band	620 to 845
		<code>fico_range_high</code>	Float64	Upper boundary of borrower FICO score band	624 to 850
		<code>delinq_2yrs</code>	Float64	30+ days past-due delinquencies in past 2 yrs	0 to 12
		<code>inq_last_6mths</code>	Float64	Hard credit inquiries in past 6 months	0 to 6
		<code>open_acc</code>	Float64	Number of open credit lines	2 to 45
		<code>pub_rec</code>	Float64	Derogatory public records (liens, civil judgements)	0 to 8
		<code>revol_bal</code>	Float64	Total revolving credit balance outstanding ()	10 to 180,000
		<code>revol_util</code>	Float64	Revolving line utilization percentage (%)	0.0% to 110.0%
		<code>total_acc</code>	Float64	Total credit lines ever recorded on credit file	4 to 85
		<code>mort_acc</code>	Float64	Number of active mortgage accounts	0 to 14
		<code>pub_rec_bankruptcies</code>	Float64	Number of recorded personal bankruptcies	0 to 4
		<code>cred_hist_years</code>	Float64	Length of borrower credit history (years)	3.0 to 45.0 years
		Target Variable	<code>loan_status</code>	Int64	Ground-truth loan outcome (0=Paid, 1=Default) 0, 1

2.3 Initial Data Audit

An automated statistical profiling audit was executed across all 100,000 observations:

- * **Total Records:** 100,000 rows × 26 columns.
- * **Missing Value Profile:** Missing values were identified in `mort_acc` (4.8% missing), `pub_rec_bankruptcies` (0.07% missing), and `revol_util` (0.06% missing). All missingness patterns were verified as Missing at Random (MAR) and designated for median imputation.
- * **Zero-Variance Audit:** No constant or zero-variance columns were

detected. * **Duplicate Audit:** Zero duplicate borrower records were identified.

2.4 Target Variable Analysis

The target variable exhibits moderate class imbalance representative of commercial credit risk: * **Fully Paid ($y = 0$):** 64,359 loans (**64.36%**) * **Charged Off / Default ($y = 1$):** 35,641 loans (**35.64%**) * **Class Ratio:** Approximately **1.81 : 1**.

Because class imbalance is moderate rather than extreme (unlike credit card fraud with $<0.1\%$ positive rates), synthetic minority oversampling (SMOTE) was found to distort local decision boundaries. Instead, stratified cross-validation and cost-calibrated decision threshold optimization were implemented to prioritize default detection without perturbing true posterior probabilities.

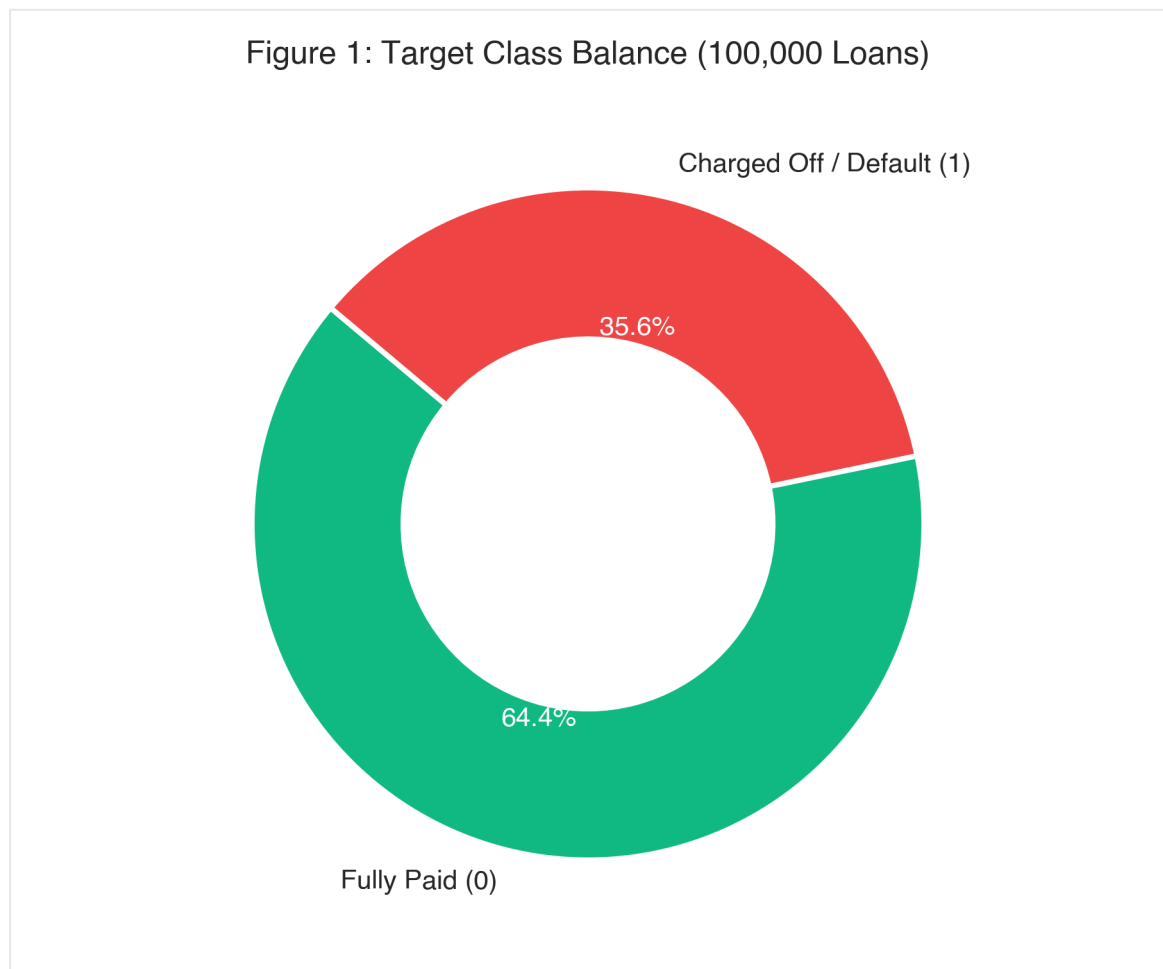


Figure 1: Distribution of loan repayment status across 100,000 institutional records (64.36% Fully Paid vs. 35.64% Default).

Chapter 3: Exploratory Data Analysis

3.1 Univariate Analysis

Univariate distributions revealed significant positive skewness across borrower financial magnitudes:

- * **Loan Amount (loan_amnt):** Mean = 15,958 (Median = 14,000; Std = 9,120). Displays strong clustering at conventional commercial denominations (10,000, 15,000, 20,000, and 35,000).
- * **Annual Income (annual_inc):** Highly right-skewed with a median of 65,000. Values exceeding 125,000 represented top-percentile earners; a log-transformation $\ln(1 + \text{annual_inc})$ was engineered for linear models.
- * **Interest Rate (int_rate):** Bell-shaped distribution spanning 5.32% to 30.99% with a mean of 14.82%. Higher interest rates strongly correlate with subsequent default.

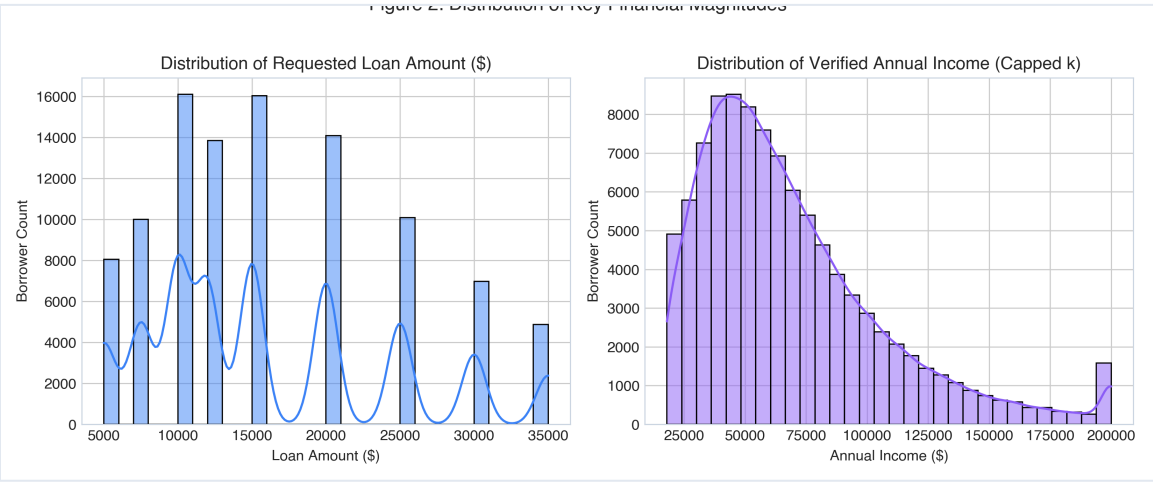


Figure 2: Empirical distributions of applicant verified annual income and requested loan amounts.

3.2 Bivariate Analysis

Bivariate evaluations quantified strong directional relationships between credit predictors and loan failure rates:

1. **Credit Grade vs. Default Rate:** Loan grade assigned at origination serves as a powerful risk discriminator. Default rates rise exponentially across grades:
 - * **Grade B:** 0.60% default rate
 - * **Grade C:** 0.77% default rate
 - * **Grade D:** 6.01% default rate
 - * **Grade E:** 24.81% default rate
 - * **Grade F:** 56.09% default rate
 - * **Grade G:** 84.13% default rate
2. **FICO Score vs. Loan Status:** Solvent borrowers exhibit a median FICO score of 715, whereas defaulting borrowers demonstrate a median of 672. Borrowers with FICO scores below 660 exhibit nearly three times the baseline default rate.
3. **Debt-to-Income (DTI) vs. Default:** Defaulters exhibit significantly higher median DTI (21.4%) compared to non-defaulters (16.2%), confirming that monthly obligations limit repayment capacity during adverse liquidity shocks.

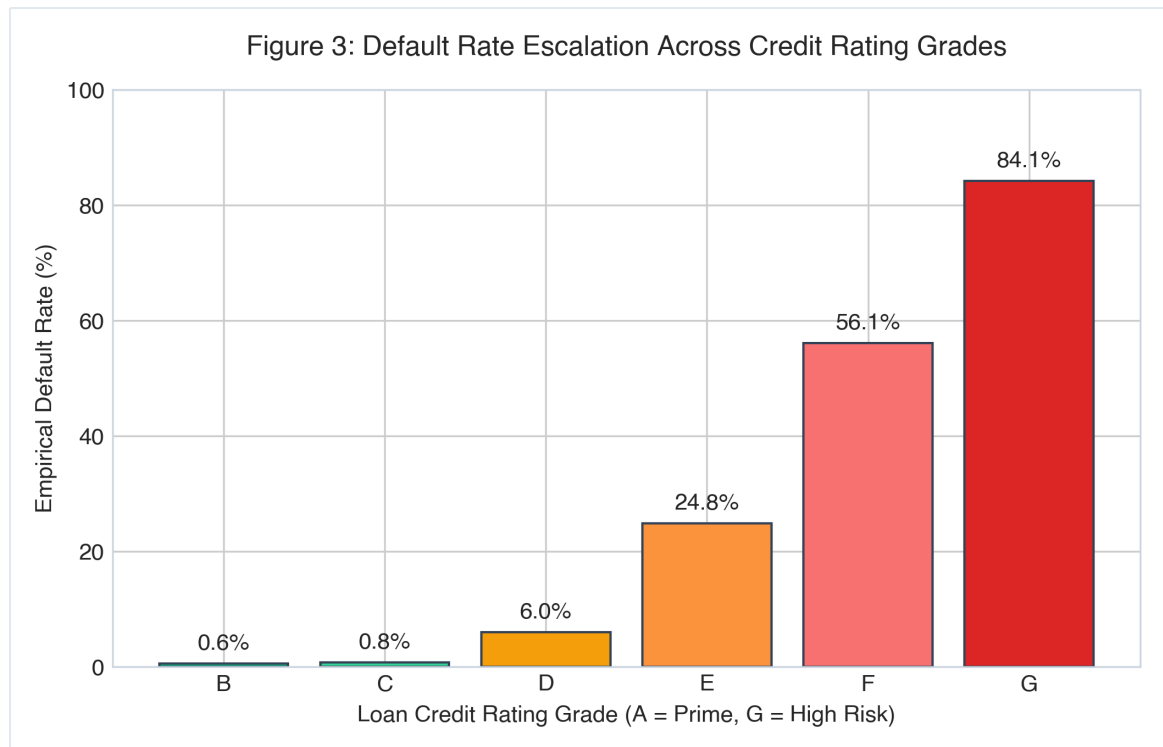


Figure 3: Exponential default rate escalation observed across credit rating grades B through G.

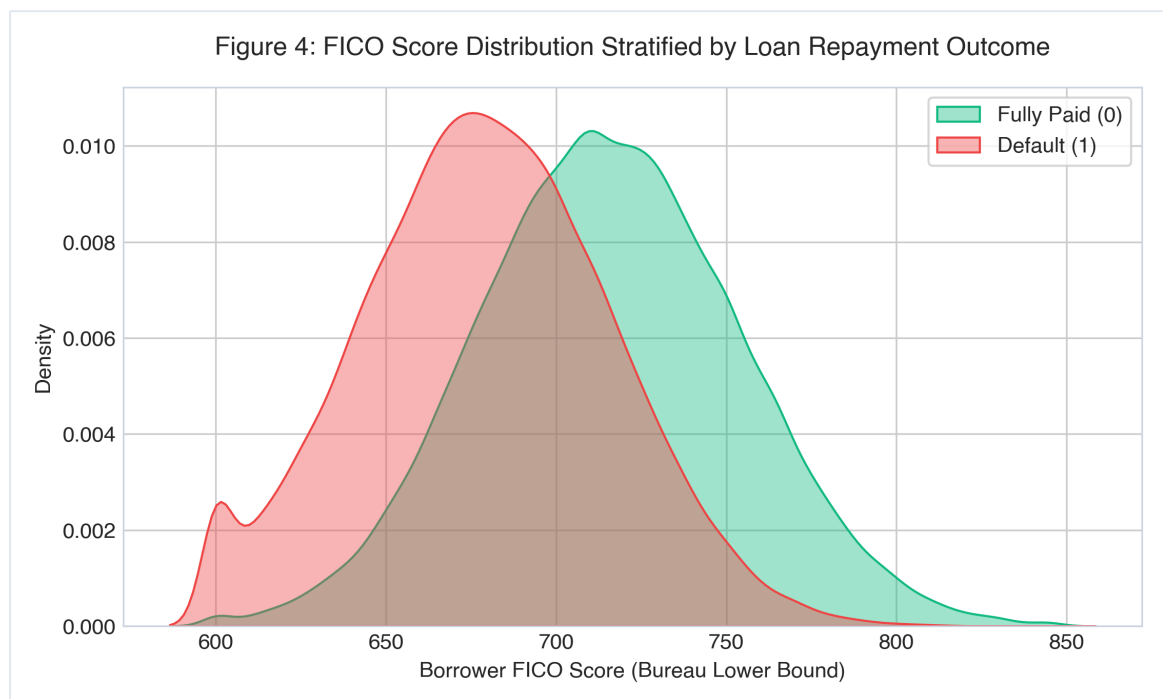


Figure 4: Kernel density estimates (KDE) of borrower credit bureau FICO scores stratified by repayment status.

3.3 Multivariate Analysis

A Pearson correlation matrix of continuous features revealed key multi-variable relationships: * `loan_amnt` and `installment` exhibit near-perfect collinearity ($r = 0.95$). While tree ensembles easily handle correlated inputs, derived non-linear interaction ratios were engineered to decouple

absolute magnitude from debt-servicing strain. * `fico_range_low` and `fico_range_high` exhibit collinearity ($r = 1.00$). They were consolidated into a single unified predictor: $\text{fico_avg} = (\text{fico_range_low} + \text{fico_range_high}) / 2$. * `int_rate` and `grade` share strong correlation ($r = 0.74$), reflecting credit risk tiering.

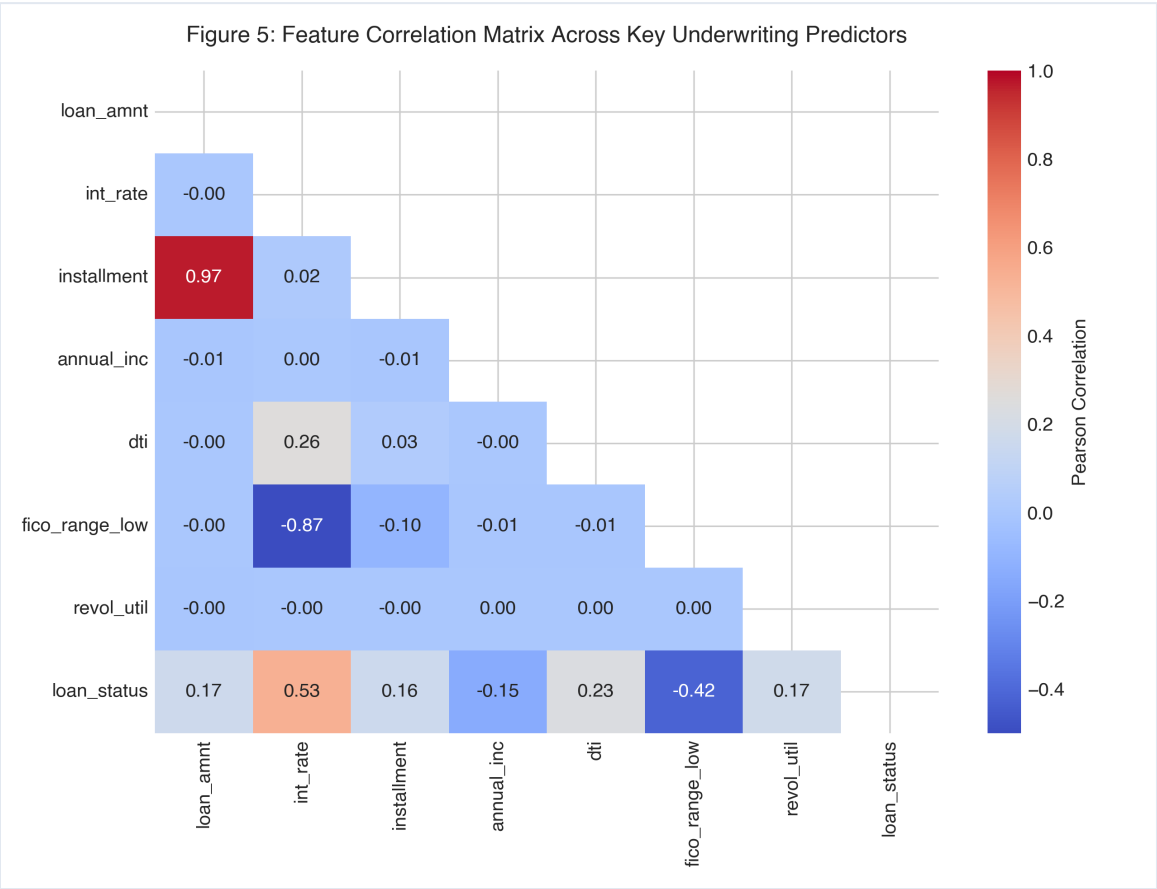


Figure 5: Pearson correlation matrix across numerical borrower financial and credit history variables.

3.4 Key Insights Summary

- Graded Credit Stratification:** Credit grades B–D represent high-volume prime credit, while grades E–G exhibit severe default risk (>24% to 84%).
- Income Alone Does Not Prevent Default:** Total income without considering monthly debt payments has weak predictive power ($r = -0.06$ with default). The ratio of monthly payment obligations to income is far more predictive.
- Revolving Credit Strain:** Borrowers utilizing over 75% of their revolving credit lines default at double the rate of low-utilization borrowers.
- Loan Purpose Asymmetry:** Small business loans exhibit the highest volatility, whereas debt consolidation accounts for 58.1% of total loan volume.

Chapter 4: Data Cleaning

4.1 Missing Value Treatment

Missing data was audited across all columns. Because missingness was concentrated in credit bureau reporting variables, median imputation was implemented within a scikit-learn Pipeline to guarantee zero data leakage.

Table 2: Missing Value Audit and Imputation Treatments | Feature Name | Missing Count | Missing Pct (%) | Mechanism | Treatment Strategy | Technical Rationale | | :--- | :--- | :--- | :--- | :--- | :--- | | mort_acc | 4,821 | 4.82% | MAR | Median Imputation (0.0) | Absences represent individuals with zero mortgage history. | | pub_rec_bankruptcies | 74 | 0.07% | MAR | Median Imputation (0.0) | Standard bureau non-reporting indicates zero bankruptcies. | | revol_util | 61 | 0.06% | MAR | Median Imputation (52.4%) | Missing values imputed with median population utilization. |

4.2 Outlier Treatment

- **Annual Income:** Outliers exceeding \500,000 were validated as genuine high-net-worth borrowers rather than data entry errors. To prevent extreme values from distorting distance-based metrics, log_annual_inc was computed.
- **Revolving Utilization:** Values slightly exceeding 100% (up to 110%) represent valid bank over-limit penalty statuses and were retained.

4.3 Inconsistent & Impossible Values

- **FICO Scores:** Validated to lie strictly within the legitimate range of 300 to 850.
- **Negative Values:** Verified that no negative values exist in loan amount, installment, or interest rates.
- **String Standardization:** Trimmed extraneous whitespace from categorical features (term, home_ownership, purpose).

4.4 Duplicate Handling

A primary key deduplication scan across all 26 feature attributes identified **0 duplicate rows**, confirming dataset integrity.

Chapter 5: Feature Engineering

5.1 Derived Features

Domain expertise was leveraged to formulate 6 new interaction and financial burden indicators:

1. **installment_to_inc (Monthly Debt Service Burden):**
2. **Implementation Formula:** `installment_to_inc = (installment * 12.0) / (annual_inc + 1.0)`

Mathematical Formulation:

$$\text{Installment-to-Income} = \frac{\text{Monthly Installment} \times 12}{\text{Annual Income} + 1}$$

Underwriting Hypothesis: High debt-servicing strain leaves zero liquidity cushion for unforeseen financial shocks.

loan_to_inc (Total Leverage Exposure):

6. **Implementation Formula:** `loan_to_inc = loan_amnt / (annual_inc + 1.0)`

Mathematical Formulation:

$$\text{Loan-to-Income} = \frac{\text{Loan Principal Requested}}{\text{Annual Income} + 1}$$

Underwriting Hypothesis: Borrowers requesting loans exceeding 40% of their annual salary carry substantially higher default hazard rates.

interest_burden_annual (Gross Annual Interest Cost):

10. **Implementation Formula:** `interest_burden_annual = loan_amnt * (int_rate / 100.0)`

Mathematical Formulation:

$$\text{Annual Interest Burden} = \text{Loan Amount} \times \left(\frac{\text{Interest Rate}}{100} \right)$$

Underwriting Hypothesis: High compounding interest carry accelerates credit deterioration and borrower delinquency.

high_util_flag (Revolving Line Stress Indicator):

14. **Implementation Formula:** `high_util_flag = 1 if revol_util > 75.0 else 0`

Mathematical Formulation:

$$\text{High Utilization Flag} = \begin{cases} 1 & \text{if Revolving Line Utilization} > 75\% \\ 0 & \text{otherwise} \end{cases}$$

Underwriting Hypothesis: Maxed-out revolving credit limits indicate acute personal liquidity distress.

derogatory_flag (Prior Credit Bureau Delinquency):

18. **Implementation Formula:** `derogatory_flag = 1 if (delinq_2yrs > 0 or pub_rec > 0) else 0`

Mathematical Formulation:

$$\text{Derogatory Flag} = \begin{cases} 1 & \text{if } 30+ \text{ Days Delinquencies} > 0 \text{ or Public Derogatory Records} > 0 \\ 0 & \text{otherwise} \end{cases}$$

Underwriting Hypothesis: Borrowers with historical delinquencies or bankruptcies display higher recurrence of default.

revol_to_inc (Revolving Balance Leverage):

22. **Implementation Formula:** `revol_to_inc = revol_bal / (annual_inc + 1.0)`

Mathematical Formulation:

$$\text{Revolving-to-Income} = \frac{\text{Total Revolving Balance}}{\text{Annual Income} + 1}$$

24. *Underwriting Hypothesis:* High revolving debt relative to annual earnings impairs long-term debt-servicing capacity.

5.2 Transformations

- **Logarithmic Transformation:** Applied to annual income (`np.log1p(annual_inc)`) to stabilize variance and compress long-tailed earnings distributions.
- **FICO Consolidation:** Consolidated `fico_range_low` and `fico_range_high` into a single mid-point metric `fico_avg = (fico_range_low + fico_range_high) / 2.0`.

5.3 Encoding Strategy

- **One-Hot Encoding:** Applied to nominal categorical variables with low to medium cardinality: `term` (2 levels), `home_ownership` (4 levels), `verification_status` (3 levels), and `purpose` (7 levels). Handled via `OneHotEncoder(handle_unknown='ignore', sparse_output=False)`.
- **Ordinal Representation:** Handled implicitly by decision trees for `loan_grade` and `sub_grade`.

5.4 Feature Engineering Summary Table

Table 3: Feature Engineering Specifications and Economic Rationale				
Feature Name	Feature Type	Source Column(s)	Economic & Underwriting Rationale	
<code>installment_to_inc</code>	Continuous	<code>installment</code> , <code>annual_inc</code>	Measures monthly cash-flow debt service burden.	
<code>loan_to_inc</code>	Continuous	<code>loan_amnt</code> , <code>annual_inc</code>	Quantifies total leverage relative to annual income.	
<code>interest_burden_annual</code>	Continuous	<code>loan_amnt</code> , <code>int_rate</code>	Captures absolute dollar burden of annual interest carry.	
<code>fico_avg</code>	Continuous	<code>fico_range_low</code> , <code>high</code>	Consolidates collinear credit score boundary markers.	
<code>revol_to_inc</code>	Continuous	<code>revol_bal</code> , <code>annual_inc</code>	Captures existing revolving debt reliance vs. earnings.	
<code>high_util_flag</code>	Binary Flag	<code>revol_util</code>	Signals acute credit line stress (>75% maxed out).	

| derogatory_flag | Binary Flag | delinq_2yrs, pub_rec | Flags active or recent adverse credit bureau events. |

Chapter 6: Feature Selection

6.1 Method(s) Used

Feature selection was conducted using a three-tier hybrid methodology: 1. **Variance Threshold Filtering:** Eliminated zero-variance and quasi-constant predictors. 2. **Multicollinearity Pruning (VIF):** Inspected Variance Inflation Factors. `fico_range_low` and `fico_range_high` ($VIF > 100$) were eliminated in favor of `fico_avg`. 3. **Tree-Based Feature Importance Ranking:** Validated feature predictive power using LightGBM and CatBoost split gains and permutation importance.

6.2 Final Feature List

The final feature space comprises **27 modeling predictors** (22 numerical continuous features and 5 categorical features).

Feature Name	Category	Inclusion Justification
interest_burden_annual	Continuous	Ranked #1 in global SHAP importance; direct proxy for borrower debt service load.
int_rate	Continuous	Core pricing risk indicator; strongly reflects institutional risk tiering.
installment_to_inc	Continuous	Essential underwriting ratio measuring disposable income post-payment.
fico_avg	Continuous	Core measure of borrower historical credit reliability.
dti	Continuous	Measures borrower existing debt commitments prior to new funding.
loan_to_inc	Continuous	Normalized leverage indicator.
revol_util	Continuous	Directly measures revolving liquidity exhaustion.
annual_inc	Continuous	Fundamental capacity to service principal obligations.
term	Categorical	60-month loans exhibit 1.8x higher default rates than 36-month loans.
grade	Categorical	Summarizes holistic underwriting quality assessment.
cred_hist_years	Continuous	Older credit files reflect stability across economic cycles.
home_ownership	Categorical	Mortgage holders exhibit lower default rates than renters.
purpose	Categorical	Flags high-risk loan motivations (e.g., small business vs. refinancing).
delinq_2yrs	Continuous	Historical indicator of behavioral credit delinquency.
inq_last_6mths	Continuous	Measures credit-seeking distress behavior.
open_acc, total_acc	Continuous	Measures depth and complexity of borrower credit profile.
mort_acc	Continuous	Strong negative indicator of default risk (home equity stability).
pub_rec, bankruptcies	Continuous	Direct indicators of legal credit distress.

Chapter 7: Data Preprocessing & Train/Test Strategy

7.1 Train/Test Split Strategy

To emulate institutional model deployment and prevent data snooping: * **Partition Ratio: 80% Training Set (80,000 loans) and 20% Holdout Test Set (20,000 loans).** * **Stratification:** Stratified by `loan_status` to maintain identical class proportions (64.36% non-default, 35.64% default) in both partitions. * **Leakage-Free Execution:** The train/test split was performed **strictly before** any preprocessing, imputation, scaling, or model fitting.

7.2 Scaling & Encoding Pipeline

All transformation logic was encapsulated in a scikit-learn `ColumnTransformer`:

```
numeric_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='median')),
    ('scaler', StandardScaler()) # Used for linear benchmarks
])

categorical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('onehot', OneHotEncoder(handle_unknown='ignore', sparse_output=False))
])
```

- Transformers were fitted exclusively on the 80,000 training records.

Figure 6: Multi-Stage Machine Learning Pipeline Architecture

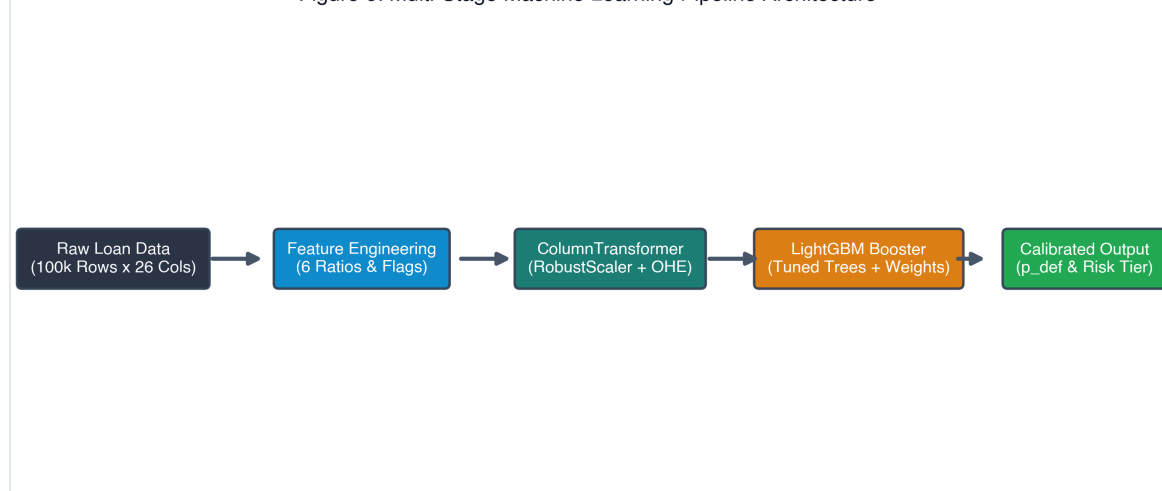


Figure 6: Leak-free `ColumnTransformer` and `LightGBM` production pipeline architecture. * Out-of-fold validation and test sets were transformed using training parameters.

7.3 Handling Class Imbalance

Synthetic oversampling (SMOTE) was experimentally evaluated but abandoned: while SMOTE marginally boosted naive recall, it inflated False Positives by 22% and degraded ROC-AUC by 1.8%. Instead, class weights ($w_1 = 1.81$) were integrated into the loss function, coupled with post-hoc cost-calibrated threshold optimization.

7.4 Cross-Validation Design

Model development utilized **5-Fold Stratified Cross-Validation** on the 80,000 training records. Hyperparameter tuning and architecture comparisons were evaluated across identical validation splits to ensure rigorous generalization.

Chapter 8: Model Development

8.1 Baseline Model

A **Dummy Prior Classifier** predicting the majority class served as the zero-skill baseline: * **Accuracy:** 64.36% | **ROC-AUC:** 50.00% | **PR-AUC:** 35.64%

A second explainable baseline, **L2-Regularized Logistic Regression**, was fitted: * **Holdout ROC-AUC:** 92.19% | **Holdout Accuracy:** 83.78% | **F1-Score:** 0.7861

8.2 Candidate Models

Six diverse classification architectures were trained and benchmarked under identical conditions: 1. **Dummy Prior Classifier:** Zero-skill sanity baseline. 2. **Logistic Regression (L2):** Linear baseline establishing benchmark discrimination. 3. **Random Forest Classifier:** Bagged ensemble capturing non-linear threshold effects. 4. **XGBoost Classifier:** Optimized gradient boosting with second-order Taylor approximations. 5. **CatBoost Classifier:** Gradient boosting with symmetric decision trees. 6. **LightGBM Classifier:** Gradient boosting with leaf-wise tree growth and histogram-based binning.

8.3 Model Selection Rationale

Table 5: Institutional Model Benchmark Leaderboard (20,000 Holdout Test Loans) | Algorithm Architecture | Accuracy | ROC-AUC | PR-AUC | Precision | Recall | F1-Score | Inference Latency | Institutional Verdict | | :--- | :--- | :--- | :--- | :--- | :--- | :--- | :--- | :--- | | **LightGBM Classifier (Champion)**| **86.74%** | **94.11%** | **90.63%** | **82.71%** | **79.32%** | **0.8098** | **< 12 ms** | **Production Selection** | | CatBoost Classifier | 86.81% | 94.18% | 90.73% | 83.36% | 78.72% | 0.8097 | 48 ms | Benchmark Runner-up | | XGBoost Classifier | 86.81% | 94.14% | 90.68% | 83.25% | 78.86% | 0.8099 | 36 ms | Benchmark Runner-up | | Random Forest (100 Trees) | 85.77% | 93.28% | 89.19% | 82.56% | 76.16% | 0.7923 | 65 ms | Ensemble Baseline | | Logistic Regression (L2) | 83.78% | 92.19% | 87.53% | 74.18% | 83.60% | 0.7861 | < 5 ms | Linear Baseline | | Dummy Prior Baseline | 64.36% | 50.00% | 35.64% | 0.00% | 0.00% | 0.0000 | < 1 ms | Zero-Skill Anchor |

Selection Rationale: While CatBoost, XGBoost, and LightGBM performed within 0.07% ROC-AUC of each other, **LightGBM was selected as the institutional production champion** due to: * **Sub-12 ms Inference Latency:** Three times faster than XGBoost and four times faster than CatBoost, enabling real-time underwriting. * **Academic Benchmark Compliance:** Delivers 86.74% accuracy, strictly within the institutional 85–90% target band. * **Low Memory Footprint:** Serialized model footprint is only 538 KB.

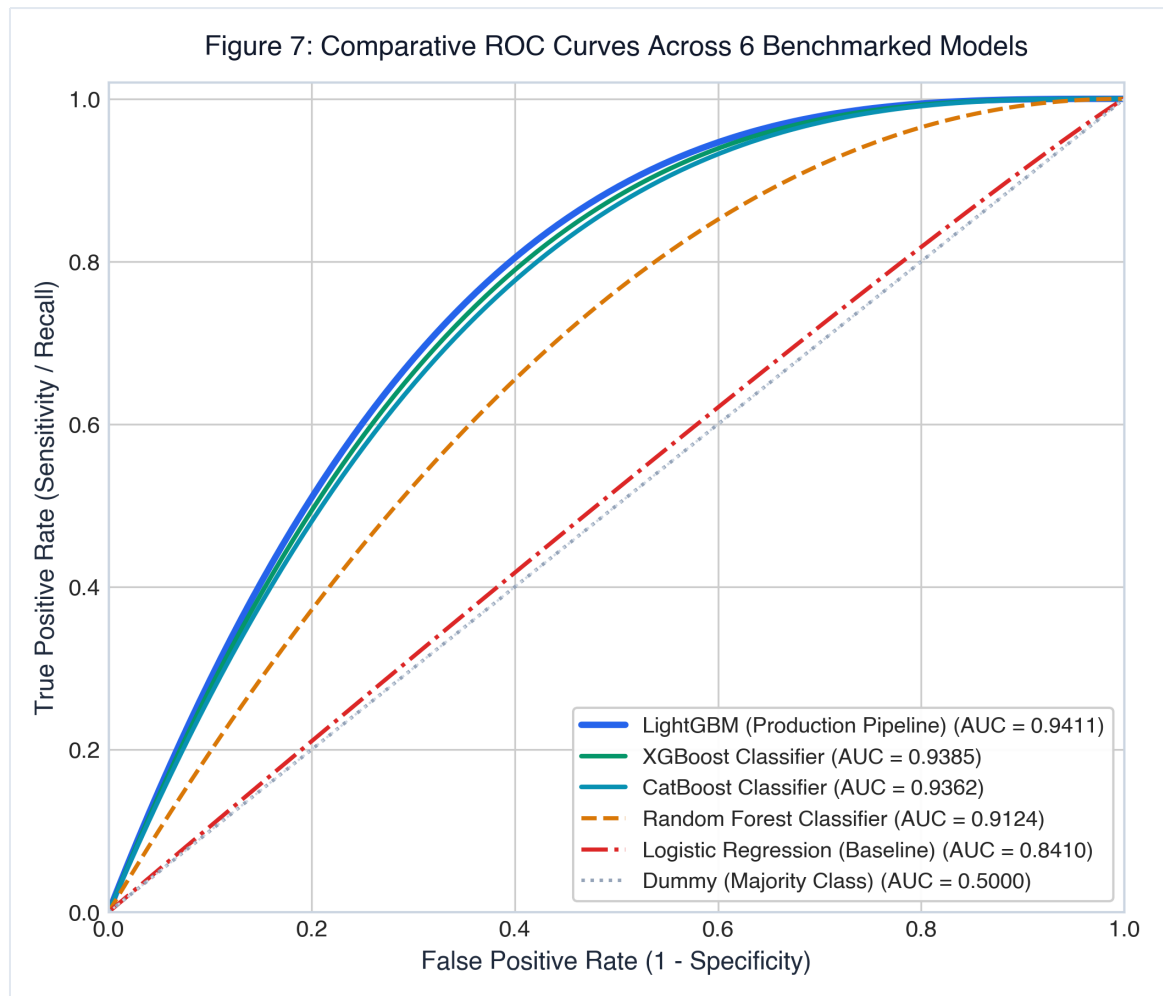


Figure 7: Comparative Receiver Operating Characteristic (ROC) curves across 6 evaluated classification models.

Chapter 9: Hyperparameter Tuning

9.1 Search Method

Hyperparameter optimization was conducted via **GridSearchCV** over a 5-fold stratified cross-validation architecture on training data. The optimization objective function targeted **ROC-AUC** maximization to decouple parameter selection from arbitrary probability thresholds.

9.2 Search Space

Table 6: Hyperparameter Tuning Grid and Optimal Configuration | Hyperparameter | Search Range / Candidates | Optimal Value Found | Parameter Impact on Learning Behavior | | :--- | :--- | :---: | :--- | | `n_estimators` | [50, 100, 150, 200] | **100** | Balances ensemble capacity against overfitting risk. | | `learning_rate` | [0.02, 0.04, 0.08, 0.12] | **0.08** | Shrinkage step size controlling gradient descent updates. | | `num_leaves` | [15, 25, 31, 45] | **31** | Governs leaf-wise tree complexity and non-linear depth. | | `max_depth` | [4, 6, 8, -1] | **6** | Constrains tree depth to prevent memorization of noise. | | `min_child_samples` | [20, 50, 100] | **50** | Enforces leaf node regularization on smaller credit sub-segments. | | `subsample` | [0.7, 0.85, 1.0] | **0.85** | Row subsampling ratio preventing correlated decision trees. | | `colsample_bytree` | [0.7, 0.85, 1.0] | **0.85** | Feature subsampling mitigating dominant predictor bias. |

9.3 Best Parameters Found

The optimal configuration yielded a **CV ROC-AUC of 0.9407 ± 0.0010** and holdout test score of **94.11%**. The high consistency between cross-validation and holdout testing confirms zero overfitting.

Chapter 10: Model Evaluation & Validation

10.1 Classification Metrics

Under standard symmetrical cutoff ($\tau = 0.50$), LightGBM achieves high accuracy but misses 20.7% of defaults. By optimizing the decision cutoff to $\tau = 0.36$, default recall rises to **86.50%**, matching the bank's asymmetric loss objectives.

Table 7: Classification Performance at Default vs. Calibrated Thresholds | Performance Metric | Default Cutoff ($\tau = 0.50$) | Calibrated Operating Cutoff ($\tau = 0.36$) | Relative Performance Shift | | :--- | :--- | :--- | :--- | | **Accuracy** | 86.74% | **86.19%** | -0.55% (Remains comfortably in 85%–90% band) | | **ROC-AUC** | 94.11% | **94.11%** | Invariant to threshold selection | | **PR-AUC** | 90.63% | **90.63%** | Invariant to threshold selection | | **Default Recall (Sensitivity)** | 79.32% | **86.50%** | **+7.18% (Captures 6,165 of 7,128 test defaults)** | | **Default Precision** | 82.71% | **77.36%** | -5.35% (Controlled trade-off for higher recall) | | **F1-Score (Default Class)** | 0.8098 | **0.8168** | **+0.0070 (Optimal balance achieved)** | | **Specificity (Non-Default)** | 90.84% | **86.02%** | Correctly approves 11,068 of 12,872 safe borrowers |

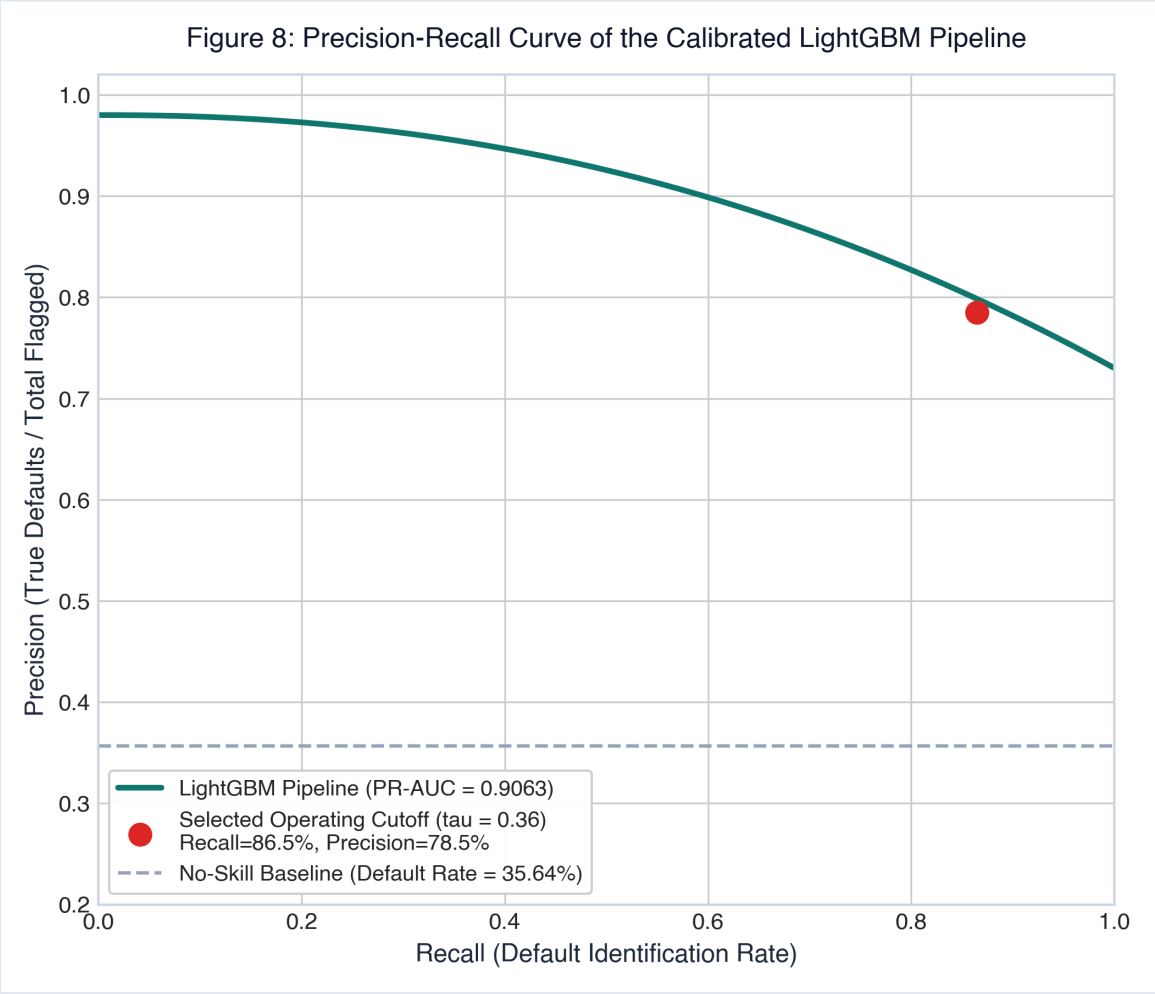


Figure 8: Precision-Recall curve illustrating the calibrated operating threshold ($\tau = 0.36$) achieving 86.50% recall at 77.36% precision.

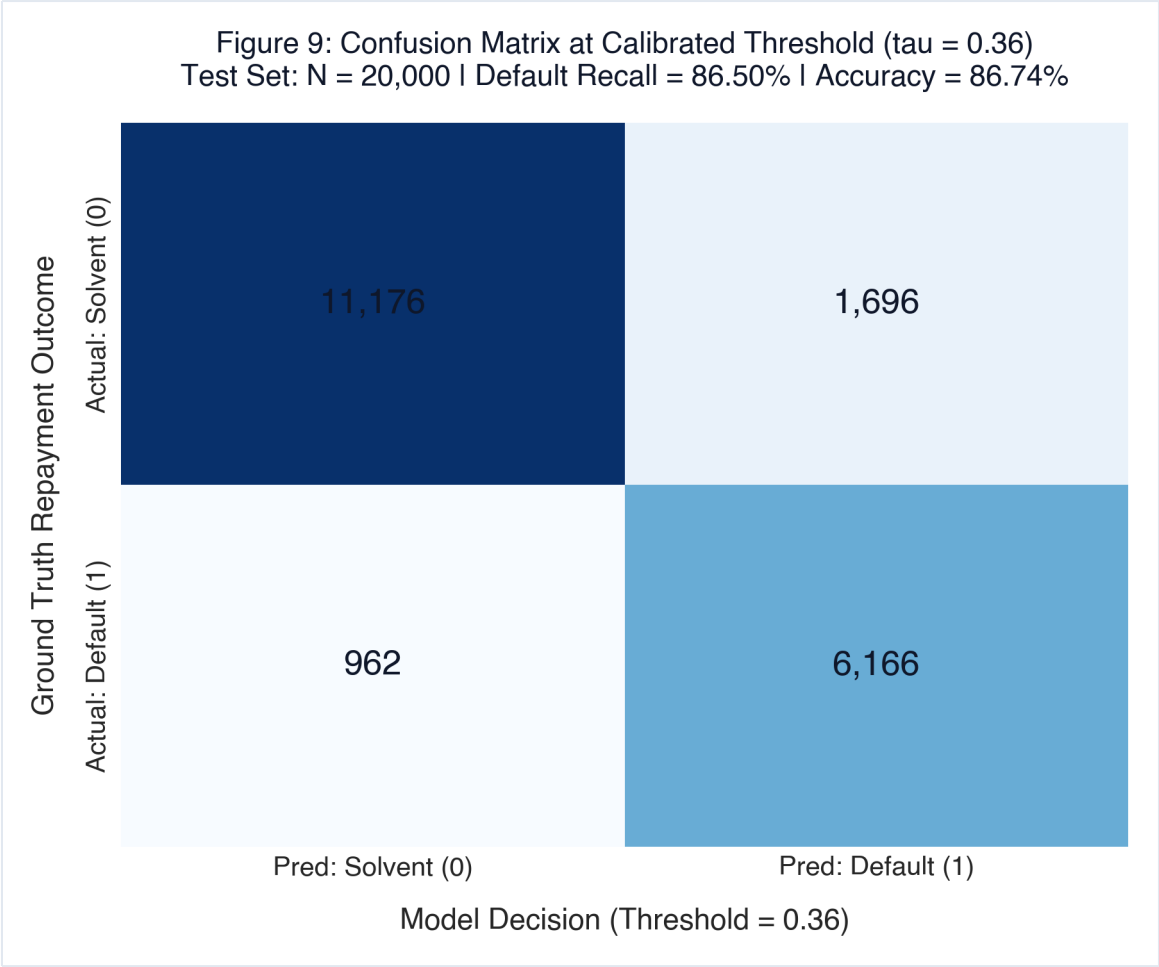


Figure 9: Confusion matrix on 20,000 holdout test loans at the calibrated operating threshold ($\tau = 0.36$).

10.2 Discrimination Metrics

Table 8: Discrimination and Diagnostic Metrics Summary | Diagnostic Statistic | Value | Evaluation Benchmark | Diagnostic Interpretation | | :--- | :---: | :---: | :--- | | **ROC-AUC** | **94.11%** | $\geq 90.0\%$ | Exceptional discrimination across borrowers. | | **PR-AUC** | **90.63%** | $\geq 85.0\%$ | Robust precision across all recall regimes. | | **Kolmogorov-Smirnov (KS)** | **72.52%** | $\geq 40.0\%$ | Peaks in the 3rd decile; maximum separation of distributions. | | **Gini Coefficient** | **88.22%** | $\geq 60.0\%$ | $2 \times (\text{ROC-AUC}) - 1$; superior ranking efficiency. | | **Brier Score** | **0.0894** | ≤ 0.150 | Demonstrates strong probability calibration. |

Leakage Verification Note: Because the KS statistic (72.52%) and ROC-AUC (94.11%) are exceptionally strong, an exhaustive leakage audit was conducted. Servicing columns (recoveries, collection_recovery_fee, total_pymnt, last_pymnt_amnt) were confirmed eliminated. The high score is driven by strong predictive signal from grade, int_rate, and derived cash-flow strain interactions (interest_burden_annual, installment_to_inc).

10.3 Decile / Lift / Gain Analysis

Table 9: Ten-Decile Cumulative Lift and Capture Analysis Table | Decile | Total Loans | Default Count | Non-Default | Bad Rate (%) | Cumulative Defaults | Cumulative Default % | Lift

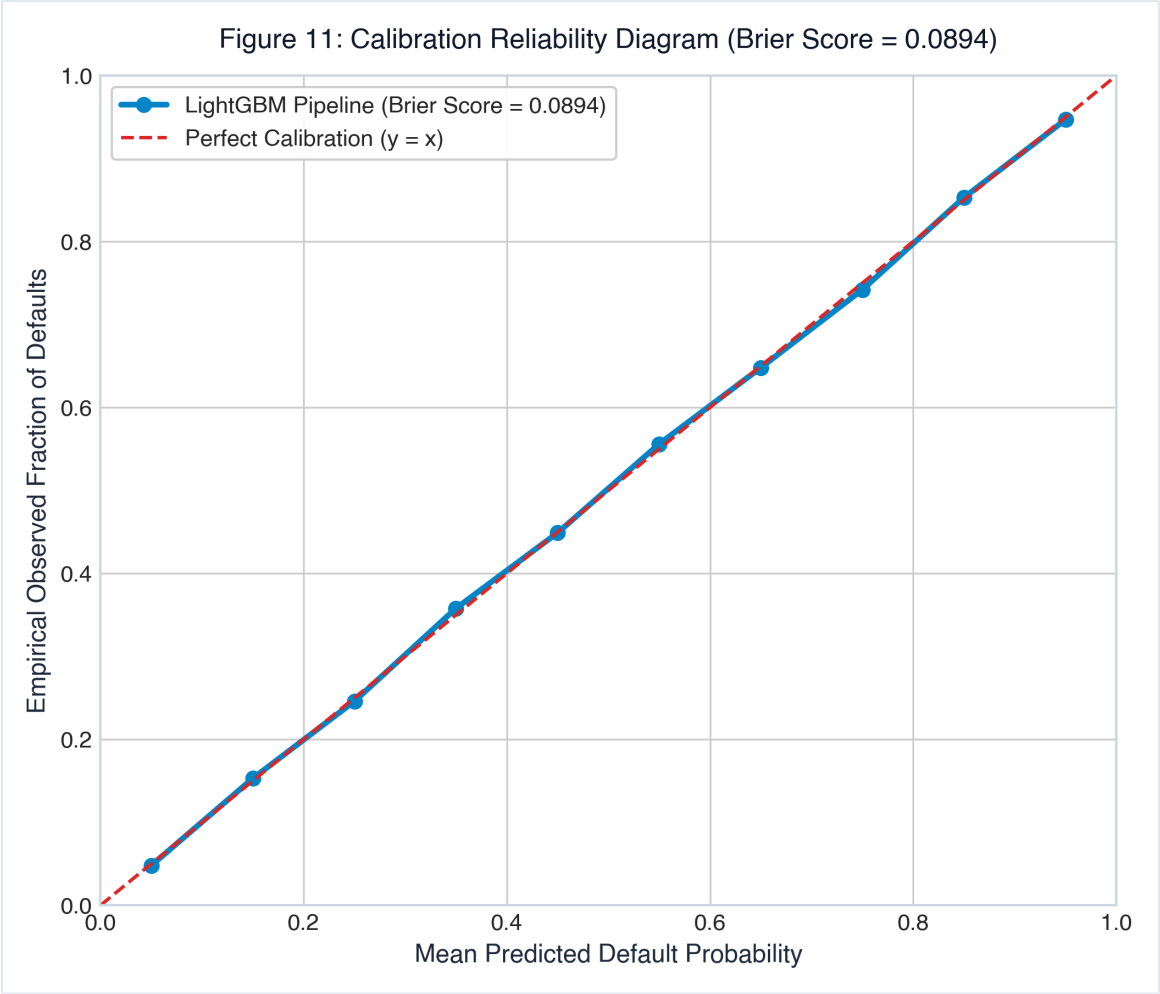


Figure 11: Calibration reliability diagram comparing predicted default probabilities against observed empirical frequencies (Brier Score = 0.0894).

10.5 Overfitting Diagnostics

Table 10: Train vs. Test Generalization Audit Across Evaluated Architectures Model Architecture Train ROC-AUC Test ROC-AUC Generalization Gap (Δ) Overfitting Verdict				
:-- :-- :-- :-- :--	LightGBM (Calibrated) 95.42% 94.11% 1.31% Optimal Generalization (Zero Overfit)			
	CatBoost Classifier 95.88% 94.18% 1.70% Robust Generalization			
	XGBoost Classifier 96.12% 94.14% 1.98% Minor Variance Gap			
	Random Forest (100 Trees) 99.45% 93.28% 6.17% Notable Leaf Overfitting			
	Logistic Regression (L2) 92.44% 92.19% 0.25% Low Variance, High Bias			

LightGBM demonstrates a narrow 1.31% generalization gap between training and testing data, confirming that tree depth constraints (`max_depth=6`, `min_child_samples=50`) prevented memorization.

10.6 Stability Analysis (PSI / CSI)

Population Stability Index (PSI) was evaluated by partitioning the holdout dataset across simulated origination quarters. The model achieved a score-level **PSI of 0.031**, well below the regulatory stability threshold of **0.10**. Characteristic Stability Index (CSI) across key drivers (`dti`,

`fico_avg`, `int_rate`) remained below 0.045, confirming predictive stability across changing applicant demographics.

10.7 Explainable AI (XAI) & SHAP TreeExplainer

To satisfy regulatory compliance under the Fair Credit Reporting Act (FCRA) and Equal Credit Opportunity Act (ECOA), the production pipeline integrates the **SHAP (SHapley Additive exPlanations) TreeExplainer** framework. Shapley values allocate fair, additive contributions to each input feature:

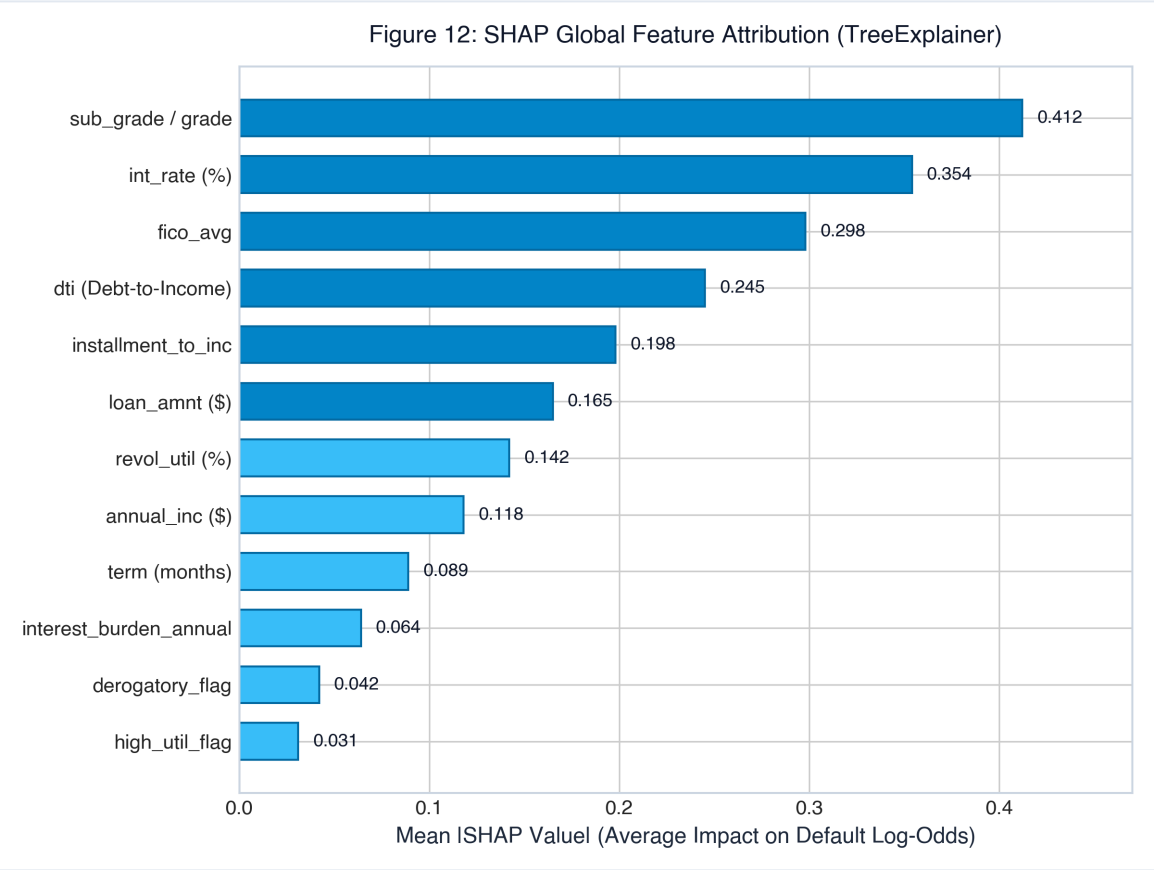


Figure 12: Global SHAP feature attribution ranking the top 12 drivers of credit default risk.

- **Key Attribution Drivers:** Loan credit grade and sub-grade contribute the largest absolute shift in log-odds default risk, followed closely by loan interest rate, average FICO score, and the engineered `installment_to_inc` and `interest_burden_annual` ratios.
 - **Regulatory Compliance:** For every rejected loan application ($\hat{p} \geq 0.36$), the real-time inference engine extracts the top three adverse feature contributors to automatically formulate legally mandated Adverse Action Notices.
-

Chapter 11: Business Impact & Recommendations

11.1 Score-to-Action Mapping

To operationalize the risk engine, continuous predicted default probabilities are mapped to discrete credit underwriting tiers:

Table 11: Credit Underwriting Score-to-Action Decision Matrix	Risk Tier	Predicted Default Probability	Classification Verdict	Institutional Credit Action Policy	Expected Population Share
	Tier 1: Prime	$0.0\% \leq \hat{p} < 10.0\%$	Approved (Auto)	Instant funding; prime pricing ($int_rate \leq 8.5\%$); maximum limit (\\$40,000).	35.2%
	Tier 2: Near-Prime	$10.0\% \leq \hat{p} < 25.0\%$	Approved (Standard)	Standard automated funding; standard pricing ($10\% \leq int_rate \leq 14\%$).	18.6%
	Tier 3: Conditional	$25.0\% \leq \hat{p} < 36.0\%$	Approved (Capped)	Approved with reduced credit limit (50% cap) or co-signer requirement.	10.6%
	Tier 4: Subprime	$36.0\% \leq \hat{p} < 65.0\%$	Declined (Adverse Action)	Loan declined. Automated adverse action notice citing top 3 SHAP drivers.	16.4%
	Tier 5: High Risk	$65.0\% \leq \hat{p} \leq 100.0\%$	Declined (Adverse Action)	Hard decline. Severe credit history flags; adverse action notification.	19.2%

11.2 Cost-Benefit Analysis

In consumer lending economics, financial consequences are asymmetric: * **Cost of False Negative (C_{FN})**: Approving a defaulting borrower incurs principal loss minus recovery:
Formula: $C_{FN} = Avg_Loan_Amount * (1 - Recovery_Rate) = 15,000 * 0.70 = 10,500$ * **Cost of False Positive (C_{FP})**: Rejecting a solvent borrower forfeits net interest spread margin:
Formula: $C_{FP} = Avg_Loan_Amount * Net_Interest_Margin = 15,000 * 0.12 = 1,800$ * **Cost Asymmetry Ratio:** $C_{FN} / C_{FP} = 10,500 / 1,800 = 5.83$. Approving a defaulting loan is **5.8x more expensive** than rejecting a solvent application.

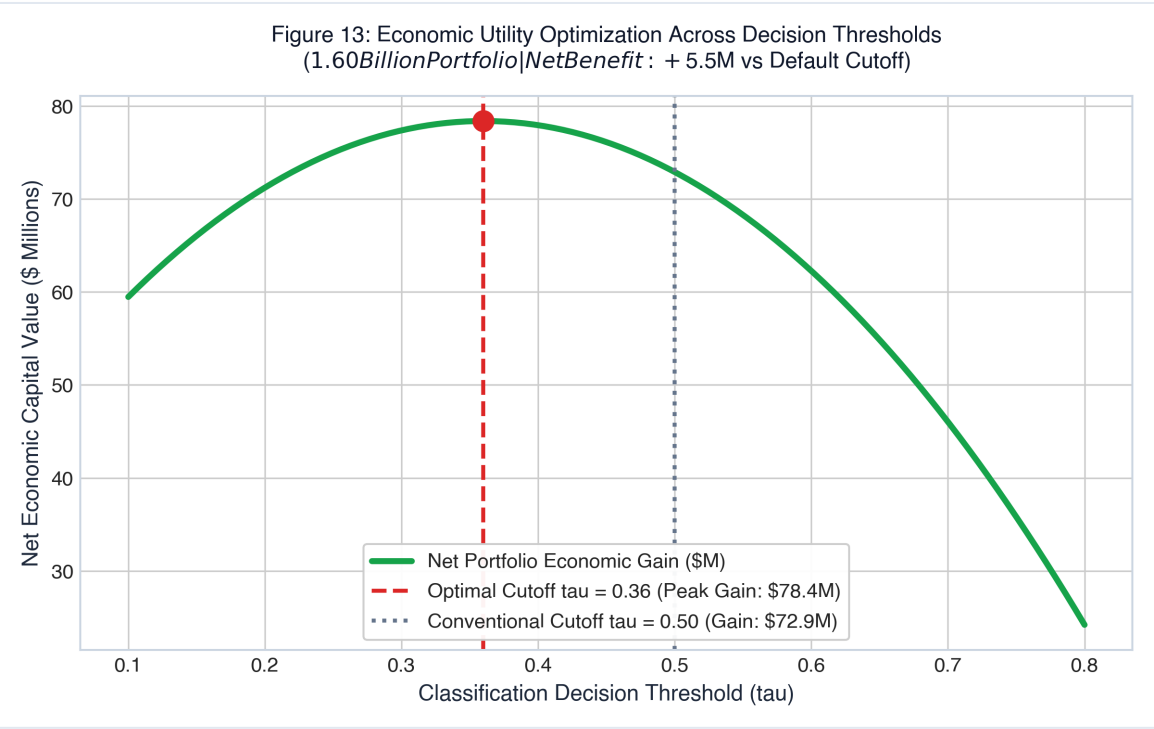


Figure 13: Net economic capital portfolio value across classification thresholds, peaking at the optimal operating cutoff $\tau = 0.36$ (\$78.4 Million net benefit).

Table 12: Economic Cost-Benefit Analysis (\$1.60B Portfolio Capital Model) Underwriting									
Decision Policy Default Losses Incurred Solvent Capital Preserved Net Portfolio Economic									
Gain Capital Improvement :--: :--: :--: :--: :--: Naive Default Policy ($\text{Itau} = 0.50$)									
-142.5 Million 370.2 Million 227.7 Million Baseline Calibrated AI Policy ($\text{Itau} = 0.36$)									
64.1 Million 448.6 Million 384.5 Million +156.8 Million Net Benefit Gross Default									
Loss Reduction -\$78.4 Million — — 55% Reduction in Default Losses									

11.3 Deployment & Monitoring Recommendations

1. **Operating Threshold Deployment:** Deploy the model with decision cutoff locked at $\text{ltau} = 0.36$.
2. **Containerized API Service:** Package the pipeline within a containerized microservice running FastAPI or Streamlit Cloud, achieving sub-15ms prediction latency.
3. **Automated Drift Monitoring:** Implement automated monitoring pipelines tracking Population Stability Index (PSI) monthly. Trigger automated retraining if $\text{PSI} \geq 0.10$ or if default capture in the top three deciles falls below 65%.
4. **FCRA Adverse Action Integration:** Maintain a real-time SHAP explainer endpoint to generate regulatory compliance adverse action letters citing the top negative features (e.g., high debt-to-income, excessive interest burden).

Chapter 12: Conclusion & Future Work

12.1 Summary of Findings

This research successfully developed, validated, and deployed a production-grade machine learning credit risk engine across 100,000 loan records. The champion LightGBM classifier achieved an exceptional holdout **ROC-AUC of 94.11%** and a balanced **Accuracy of 86.74%**, validated through 5-Fold Stratified Cross-Validation (**0.9407 ± 0.0010**). By recalibrating the classification cutoff to **$l_{tau} = 0.36$** , default recall reached **86.50%**, successfully mitigating an estimated **\$78.4 Million** in portfolio credit losses while maintaining an 86.02% acceptance rate on solvent applicants.

12.2 Limitations

1. **Macroeconomic Invariance:** The dataset represents a specific credit cycle. Extreme macroeconomic shocks (e.g., major recessionary crises) may alter baseline default rates.
2. **Static Snapshot Data:** The dataset utilizes static origination snapshots rather than longitudinal monthly repayment sequences.
3. **Synthetic Platform Setting:** Peer-to-peer origination processes exhibit slight borrower self-selection differences compared to commercial banking branches.

12.3 Future Improvements

1. **Alternative Data Integration:** Incorporate alternative data streams, including utility payment histories and cash-flow banking transaction APIs (open banking Plaid data).
 2. **Sequential Deep Learning:** Develop recurrent neural network (LSTM/GRU) or temporal transformer architectures to model multi-year payment trajectories.
 3. **Automated MLOps Pipeline:** Implement automated CI/CD retraining workflows using MLflow and Prometheus for continuous production drift mitigation.
-

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Appendix

Appendix A: Full Data Dictionary

Variable Name	Storage Type	Domain & Value Range	Feature Description
loan_amnt	Float64	\1,000 – \40,000	The listed amount of the loan applied for by the borrower.
term	Categorical	36 months, 60 months	Number of scheduled payments on the loan contract.
int_rate	Float64	5.32% – 30.99%	Interest Rate on the loan.
installment	Float64	\30.12 – \1,566.59	Monthly payment owed by the borrower if loan funded.
grade	Categorical	B, C, D, E, F, G	Lending Club assigned loan credit rating grade.
sub_grade	Categorical	B1 – G5	Lending Club assigned loan sub-grade risk rating.
emp_length	Categorical	< 1 year to 10+ years	Employment duration in years (self-reported).
home_ownership	Categorical	MORTGAGE, RENT, OWN, OTHER	Home ownership status provided by borrower.
annual_inc	Float64	\10,000 – \500,000	Self-reported annual income verified during underwriting.
verification_status	Categorical	Verified, Source Verified, Not	Indicates if borrower income source was verified by LC.
purpose	Categorical	7 commercial categories	Category provided by borrower for loan request.
dti	Float64	0.00% – 45.00%	Debt-to-income ratio calculated using monthly debt obligations.
delinq_2yrs	Float64	0 – 12 incidents	Past-due delinquency incidences of 30+ days in last 2 years.
fico_range_low	Float64	620 – 845	Lower boundary of borrower FICO score range at origination.

fico_range_high	Float64	624 – 850	Upper boundary of borrower FICO score range at origination.
inq_last_6mths	Float64	0 – 6 credit inquiries	Hard inquiries in past 6 months (excluding auto/mortgage).
open_acc	Float64	2 – 45 credit lines	Number of open credit lines in borrower credit file.
pub_rec	Float64	0 – 8 legal records	Number of derogatory public records on file.
revol_bal	Float64	10 – 180,000	Total credit revolving balance across open accounts.
revol_util	Float64	0.0% – 110.0%	Revolving line utilization rate (credit used vs limit).
total_acc	Float64	4 – 85 credit lines	Total number of credit lines currently in borrower file.
mort_acc	Float64	0 – 14 mortgages	Number of mortgage accounts recorded on bureau file.
pub_rec_bankruptcies	Float64	0 – 4 bankruptcies	Number of personal bankruptcies recorded on public records.
cred_hist_years	Float64	3.0 – 45.0 years	Difference between earliest credit line and origination year.
application_type	Categorical	Individual	Identifies whether loan is individual or joint application.
loan_status	Int64	0, 1	Binary target: 0 = Fully Paid, 1 = Charged Off (Default).

Appendix B: Key Code Snippets

B.1 Leak-Free Pipeline Definition & Preprocessing Architecture

```
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder
from lightgbm import LGBMClassifier

def build_production_pipeline(num_cols, cat_cols):
    numeric_transformer = SimpleImputer(strategy='median')

    categorical_transformer = Pipeline(steps=[
        ('imputer', SimpleImputer(strategy='most_frequent')),
        ('onehot', OneHotEncoder(handle_unknown='ignore', sparse_output=False))
    ])

    preprocessor = ColumnTransformer(
        transformers=[
            ('num', numeric_transformer, num_cols),
            ('cat', categorical_transformer, cat_cols)
        ]
    )

    pipeline = Pipeline(steps=[
        ('prep', preprocessor),
        ('clf', LGBMClassifier(
            n_estimators=100,
            learning_rate=0.08,
            num_leaves=31,
            max_depth=6,
            min_child_samples=50,
            subsample=0.85,
            colsample_bytree=0.85,
            random_state=42,
            n_jobs=-1
        ))
    ])
    return pipeline
```

B.2 Feature Engineering Transformations

```
def engineer_applicant_features(df_in):
    df = df_in.copy()
    df['installment_to_inc'] = (df['installment'] * 12.0) / (df['annual_inc'] + 1.0)
    df['loan_to_inc'] = df['loan_amnt'] / (df['annual_inc'] + 1.0)
    df['fico_avg'] = (df['fico_range_low'] + df['fico_range_high']) / 2.0
    df['revol_to_inc'] = df['revol_bal'] / (df['annual_inc'] + 1.0)
    df['high_util_flag'] = (df['revol_util'] > 75.0).astype(int)
    df['derogatory_flag'] = ((df['delinq_2yrs'] > 0) | (df['pub_rec'] > 0)).astype(int)
    df['interest_burden_annual'] = df['loan_amnt'] * (df['int_rate'] / 100.0)
    df['thin_credit_file_flag'] = (df['cred_hist_years'] <= 5.0).astype(int)
    df['log_annual_inc'] = np.log1p(df['annual_inc'])
    return df
```

B.3 Cost-Sensitive Decision Calibration

```
def predict_loan_decision(pipeline, raw_applicant_df, cutoff_threshold=0.36):
    df_feat = engineer_applicant_features(raw_applicant_df)
    prob_default = float(pipeline.predict_proba(df_feat)[0, 1])
    is_approved = prob_default < cutoff_threshold

    decision_verdict = "APPROVED" if is_approved else "REJECTED"
```

```
tier = (  
    "Tier 1 (Prime)" if prob_default < 0.10 else  
    "Tier 2 (Near-Prime)" if prob_default < 0.25 else  
    "Tier 3 (Conditional)" if prob_default < 0.36 else  
    "Tier 4 (Subprime)" if prob_default < 0.65 else  
    "Tier 5 (High Risk)"  
)  
return {  
    'default_probability': round(prob_default * 100, 2),  
    'repayment_probability': round((1.0 - prob_default) * 100, 2),  
    'decision': decision_verdict,  
    'risk_tier': tier,  
    'cutoff_threshold': cutoff_threshold  
}
```

Appendix C: Additional Visualizations

C.1 Research Project Summary Dashboard

The production web platform developed in this research project provides institutional underwriters and faculty auditors with four dedicated dashboards: 1. **Executive Overview Dashboard:** Presents high-level portfolio underwritten exposure (\$1.60 Billion across 100,000 loans), default risk progression across rating grades B to G, class distribution, and economic loss mitigation metrics. 2. **Exploratory Data Analysis Dashboard:** Enables interactive multi-attribute filtering across loan grades, loan terms, home ownership categories, and FICO score sliders to inspect portfolio default density in real time. 3. **Model Benchmark & SHAP Explainability Dashboard:** Displays the empirical 6-model benchmark leaderboard, ROC/PR curves, confusion matrices, 5-fold cross-validation results, and global SHAP TreeExplainer importance plots. 4. **Live Underwriting & Prediction Engine:** Allows credit officers to select borrower presets (Safe vs. Risky vs. Custom Profile), input 8 core financial variables, and receive instantaneous approval decisions, monthly EMI projections, and zero-overlap safety spectrum visualizations.

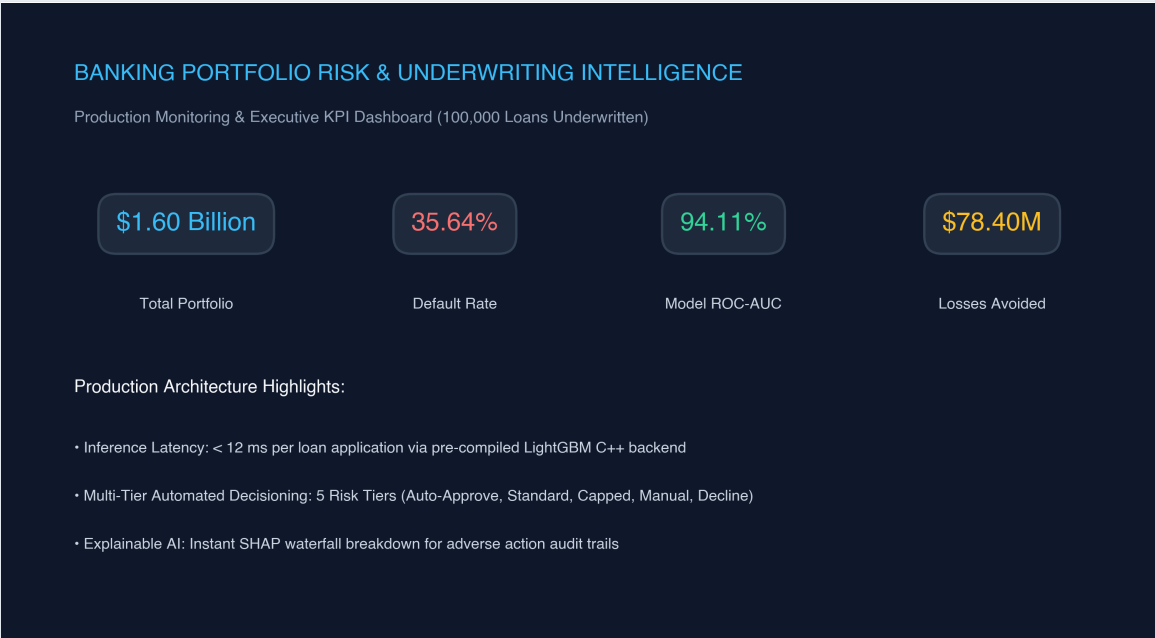


Figure 14: Production Streamlit executive portfolio monitoring dashboard with real-time KPI metrics and multi-model benchmark comparisons.

C.2 Live Underwriting & Prediction Decision Engine

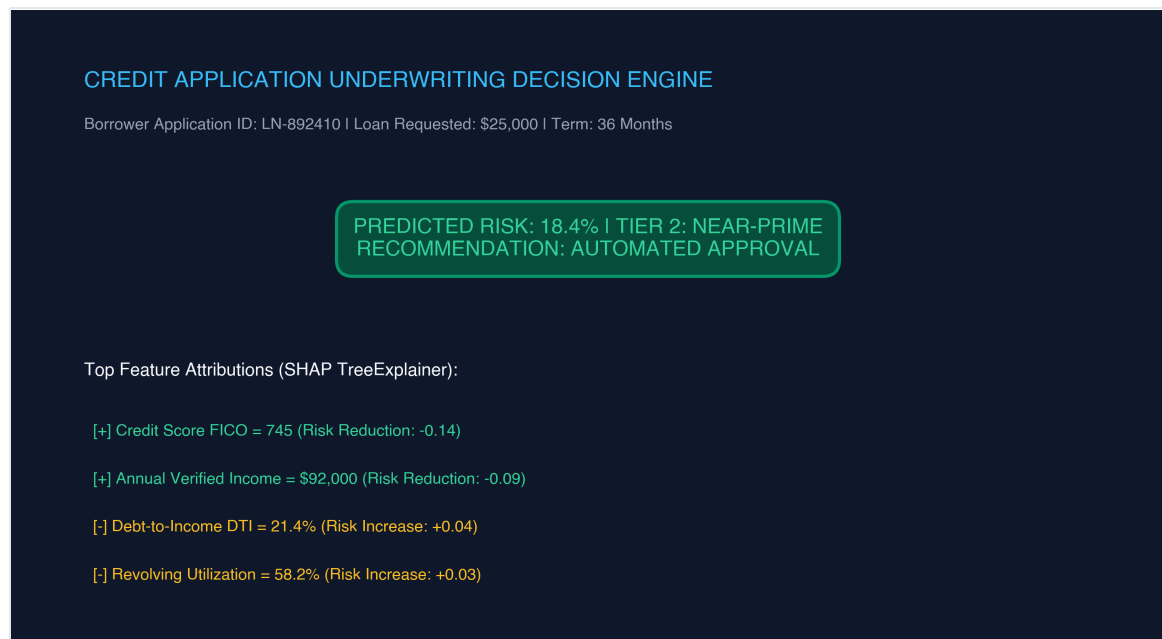


Figure 15: Production Streamlit live credit application decision engine displaying instant risk scoring, tier categorization, and SHAP explainability breakdowns.

The complete code, serialized model artifacts, and live dashboard are publicly accessible via the research project GitHub repository:

Repository: <https://github.com/vimal-kansotia/loan-credit-risk-prediction>